

Network Analysis of Questionnaire Responses

Participant–Participant and Question–Question Similarity Networks

Abstract

This report presents a correlation-based network analysis of questionnaire responses from 86 participants and 60 questionnaire items. Two complementary networks are examined: a participant–participant network, in which nodes represent participants and edges represent similarity between response profiles, and a question–question network, in which nodes represent questionnaire items and edges represent similarity between item response patterns. Spearman correlation is used as the similarity measure, and networks are evaluated over a range of absolute-correlation thresholds. Network density, connectivity, giant-component size, clustering, shortest-path structure, and diameter are compared with Erdős–Rényi (ER) random-graph baselines. The results show that both networks undergo a progressive transition from dense connectivity to fragmentation as the similarity threshold increases. The participant network retains a large connected core over a wider threshold range than the question network. In the sparse regime, both empirical networks exhibit clustering that is higher than their corresponding ER baselines, indicating that observed similarities are organized into locally cohesive structures rather than being distributed as random connections. The question network also reveals domain-level structure among Artificial Intelligence, Education, Social, and Environment items.

1 Introduction

Questionnaire data can be represented as a network in more than one way. A participant–participant network captures similarity between respondents, whereas a question–question network captures similarity between questionnaire items. These representations answer different questions: the former asks whether participants exhibit similar response profiles, while the latter asks whether questionnaire items tend to be answered similarly across the participant population.

In this analysis, pairwise Spearman correlations are used because questionnaire responses are ordinal/Likert-type measurements and Spearman’s ρ provides a rank-based measure of monotonic association. For a pair of entities i and j , an edge is retained when

$$|\rho_{ij}| \geq \tau, \tag{1}$$

where τ is the correlation threshold. The absolute value is used for network construction so that both strong positive and strong negative associations are retained; the sign of the association can subsequently be preserved in network visualization.

Rather than selecting a single threshold without examination, the network structure is evaluated across thresholds from 0.00 to 0.90. This permits assessment of threshold sensitivity and identifies the range in which the network changes from a densely connected system to a fragmented one.

2 Data and Network Construction

2.1 Participant–Participant Network

The participant network contains 86 nodes. Each node corresponds to one participant. For every pair of participants, a Spearman correlation is computed between their questionnaire response vectors. A sufficiently large absolute correlation indicates that the two participants exhibit similar response patterns and is represented by an undirected edge.

At $\tau = 0.50$, the resulting network contains 169 edges. The corresponding network visualization is shown in Figure 1.

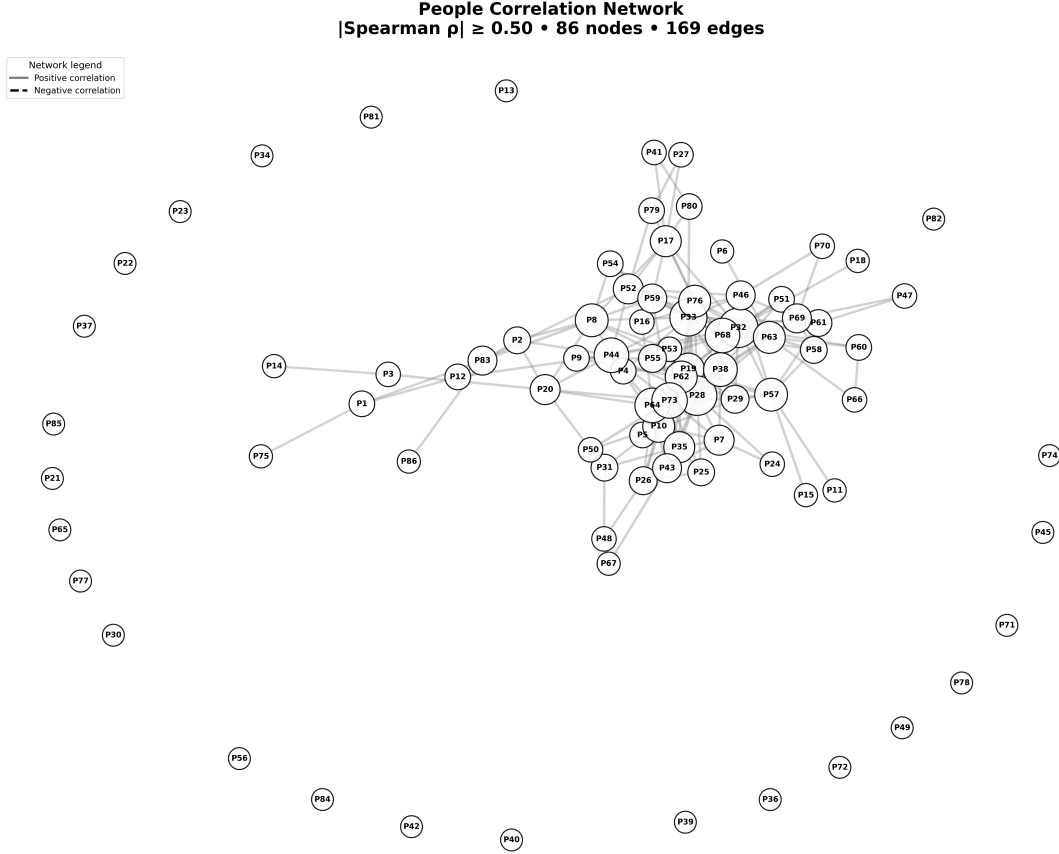


Figure 1: Participant–participant correlation network at $|\rho| \geq 0.50$. Nodes represent participants and edges represent pairwise response similarity.

2.2 Question–Question Network

The question network contains 60 nodes, corresponding to the questionnaire items. Two question nodes are connected when their response patterns across participants have an absolute Spearman correlation at least as large as the specified threshold.

At $\tau = 0.50$, only 14 edges remain, producing a sparse network. The corresponding network visualization is shown in Figure 2.

occurs at $\tau = 0.35$, where the network contains two components and the largest component still contains 85 of the 86 participants (98.8%). By $\tau = 0.40$, there are seven components and the largest component contains 79 participants (91.9%).

A more pronounced transition occurs between $\tau = 0.50$ and $\tau = 0.55$. At $\tau = 0.50$, the network contains 169 edges, has density 0.046, and its largest component contains 62 participants (72.1%). At $\tau = 0.55$, only 61 edges remain and the largest component has decreased to 41 participants (47.7%). At $\tau = 0.60$, the network contains only nine edges and the largest component contains four participants (4.7%).

Table 1: Selected topology measures for the participant network.

τ	Edges	Density	Components	Largest	Giant frac.	Clustering	Path length
0.30	1498	0.410	1	86	1.000	0.628	1.653
0.35	1033	0.283	2	85	0.988	0.510	1.858
0.40	637	0.174	7	79	0.919	0.418	1.994
0.45	348	0.095	11	76	0.884	0.278	2.371
0.50	169	0.046	25	62	0.721	0.241	2.766
0.55	61	0.017	44	41	0.477	0.117	3.577
0.60	9	0.002	77	4	0.047	0.000	1.667

3.2 Clustering relative to random networks

Clustering decreases as the threshold increases, as expected when edges are removed. However, the empirical network becomes increasingly different from the ER baseline in the sparse regime.

At $\tau = 0.40$, the observed clustering coefficient is 0.418 compared with an ER mean of 0.176. At $\tau = 0.45$, the corresponding values are 0.278 and 0.095. At $\tau = 0.50$, observed clustering is 0.241, whereas the ER mean is only 0.043. Thus, at $\tau = 0.50$, the observed clustering is approximately 5.7 times the ER mean.

This indicates that participant similarities are locally organized: participants who are similar to one another are more likely to be mutually similar than would be expected if the same number of edges were placed randomly. This is consistent with the presence of cohesive groups of participants with similar response profiles. The network topology alone, however, does not establish what substantive participant characteristics account for these groups.

3.3 Shortest paths and small-world characteristics

The average shortest-path length increases from 1.00 in the complete graph to 2.77 at $\tau = 0.50$ and 3.58 at $\tau = 0.55$. At $\tau = 0.50$, the empirical average path length is 2.77 compared with an ER mean of 3.31. Thus, the empirical network simultaneously has substantially higher clustering and shorter paths than its ER reference at this threshold.

For descriptive comparison, the commonly used small-world coefficient can be written as

$$\sigma = \frac{C/C_{\text{ER}}}{L/L_{\text{ER}}},$$

where C and L denote empirical clustering and average path length, respectively. At $\tau = 0.45$ and $\tau = 0.50$, the corresponding values from the present data are approximately 2.88 and 6.79. These values are above one, which is consistent with a small-world-like combination of high clustering and short paths. Because threshold choice and graph fragmentation affect these quantities, this should be treated as a descriptive structural result rather than a standalone claim about a unique network-generating mechanism.

3.4 Connectivity transition

The participant network shows a broad connected regime followed by a rapid loss of global connectivity. The largest component remains above 88% of participants through $\tau = 0.45$, falls to 72.1% at $\tau = 0.50$, and drops below half of the participant population at $\tau = 0.55$.

The ER baseline remains more connected than the empirical network at $\tau = 0.50$: the empirical giant fraction is 0.721 compared with an ER mean of 0.980 (SD 0.016). At $\tau = 0.55$, the empirical giant fraction is 0.477 compared with an ER mean of 0.513 (SD 0.163). The divergence at $\tau = 0.50$ suggests that the empirical network fragments more readily than a density-matched random network, consistent with non-random organization of participant similarities.

4 Question–Question Network

4.1 Threshold-dependent topology

The question network contains 60 nodes and 1,770 possible undirected edges. At $\tau = 0.00$, all possible edges are present. As the threshold increases, the network becomes sparse more rapidly than the participant network.

At $\tau = 0.25$, 501 edges remain and the network is still fully connected. At $\tau = 0.30$, only 325 edges remain, the density is 0.184, and the network has already split into four components. The largest component contains 57 of the 60 questions (95.0%). At $\tau = 0.35$, the network contains 186 edges and 12 components, with a largest component of 49 questions (81.7%). By $\tau = 0.40$, there are 93 edges, 20 components, and the largest component contains 40 questions (66.7

At $\tau = 0.50$, which corresponds to the visualization in Figure 2, only 14 edges remain. The network has 47 components and its largest component contains seven questions (11.7%). Above $\tau = 0.65$, at most one edge remains, and from $\tau = 0.70$ onward no edges remain.

Table 2: Selected topology measures for the question network.

τ	Edges	Density	Components	Largest	Giant frac.	Clustering	Path length
0.30	325	0.184	4	57	0.950	0.448	2.279
0.35	186	0.105	12	49	0.817	0.310	2.643
0.40	93	0.053	20	40	0.667	0.150	2.874
0.45	40	0.023	33	23	0.383	0.094	2.838
0.50	14	0.008	47	7	0.117	0.028	2.286
0.55	8	0.005	52	5	0.083	0.000	2.000
0.60	2	0.001	58	3	0.050	0.000	1.333

4.2 Comparison with ER networks

At low thresholds, the empirical question network is close to the ER baseline in both clustering and path length. As the graph becomes sparse, the difference becomes more pronounced.

At $\tau = 0.40$, the observed clustering coefficient is 0.150 compared with an ER mean of 0.040. The observed average path length is 2.87 compared with an ER mean of 3.51. Thus, in the connected portion of the sparse regime, the empirical network contains more local clustering and shorter paths than the corresponding random networks.

At $\tau = 0.45$, the empirical clustering is 0.094 compared with 0.0068 for the ER baseline. However, the empirical giant fraction is only 0.383 compared with an ER mean of 0.468. This indicates that the empirical network is simultaneously more locally clustered and more fragmented than the corresponding random networks.

For $\tau \geq 0.50$, the empirical question network becomes so fragmented that global path-length comparisons become less informative. The primary descriptors in this regime are therefore component counts, component sizes, and edge density.

4.3 Question-domain structure

The questionnaire items are organized into four substantive domains: Artificial Intelligence, Education, Social, and Environment. To examine whether strong question-question relationships are concentrated within or across these domains, the number of edges at $\tau = 0.30$ was tabulated by domain pair.

Table 3: Question-question edges at $|\rho| \geq 0.30$, grouped by domain pair.

Domain 1	Domain 2	Number of edges
Artificial Intelligence	Artificial Intelligence	7
Artificial Intelligence	Education	7
Artificial Intelligence	Environment	13
Artificial Intelligence	Social	14
Education	Education	26
Education	Environment	28
Education	Social	22
Social	Social	50
Environment	Social	78
Environment	Environment	80

The domain-level counts show that strong relationships are not restricted to items within the same domain. For example, the Environment-Social pair accounts for 78 edges, while the Environment-Environment and Social-Social pairs account for 80 and 50 edges, respectively. Education-Environment and Education-Social account for 28 and 22 edges, while Artificial Intelligence-Social accounts for 14 edges.

These raw edge counts should be interpreted with care because the number of possible edges differs across domain pairs. In particular, within-domain pairs and between-domain pairs have different numbers of possible connections. Consequently, raw edge counts describe the observed concentration of edges but should not by themselves be interpreted as evidence that one domain pair is intrinsically more strongly connected. A normalized domain-pair density would provide a more direct comparison.

4.4 Interpretation of the question network

The question network becomes fragmented substantially earlier than the participant network. This indicates that strong correlations between questionnaire items are relatively selective. In other words, although many items show moderate association, only a comparatively small number retain strong pairwise relationships when the threshold is increased.

The domain analysis further suggests that questionnaire constructs are not completely separable at the correlation level: strong relationships occur both within domains and between domains. This is compatible with the possibility that the questionnaire captures overlapping or related attitudinal dimensions. Establishing the psychological meaning of these relationships would require item-level content analysis and, potentially, factor-analytic or latent-variable methods.

5 Comparison of Participant and Question Networks

The two network representations exhibit qualitatively different threshold behavior. The participant network contains 86 nodes and remains fully connected through $\tau = 0.30$, with a giant component containing 72.1% of participants even at $\tau = 0.50$. The question network contains 60 nodes but becomes disconnected already at $\tau = 0.30$, and only 11.7% of its nodes remain in the largest component at $\tau = 0.50$.

This difference is visible in the number of edges at corresponding thresholds. At $\tau = 0.50$, the participant network contains 169 edges, whereas the question network contains only 14. Thus, strong similarity is much more prevalent among participant response profiles than among questionnaire-item response profiles.

The distinction is conceptually important. A participant network asks whether two respondents answer the questionnaire similarly across many items. A question network asks whether two items receive similar patterns of responses across respondents. There is no requirement that these two structures be equally connected: participants can share broad response tendencies even when individual questionnaire items do not form strongly correlated pairs.

Table 4: Side-by-side comparison at selected thresholds.

Network	τ	Edges	Density	Components	Largest	Giant frac.
People	0.30	1498	0.410	1	86	1.000
People	0.40	637	0.174	7	79	0.919
People	0.50	169	0.046	25	62	0.721
People	0.55	61	0.017	44	41	0.477
Questions	0.30	325	0.184	4	57	0.950
Questions	0.40	93	0.053	20	40	0.667
Questions	0.50	14	0.008	47	7	0.117

6 Discussion

The threshold analysis demonstrates that correlation-based questionnaire networks should not be interpreted from a single arbitrary threshold. At low thresholds, both networks are dense because even moderate correlations generate many edges. As the threshold increases, weaker associations disappear and the underlying connectivity structure becomes visible.

The participant network exhibits a comparatively robust connected core. Even after substantial edge removal, a large fraction of participants remains connected through strong similarities. At the same time, its clustering is substantially higher than the ER expectation in the sparse regime. This combination suggests that participants with similar response profiles tend to form locally cohesive groups.

The question network behaves differently. Strong item–item relationships are much more selective, leading to fragmentation at lower thresholds. The domain-level analysis shows that strong associations occur both within and across substantive domains, suggesting that the questionnaire dimensions may overlap rather than forming four strictly isolated blocks.

An important distinction is between *network structure* and *substantive interpretation*. High clustering demonstrates that similar nodes tend to be connected to one another; it does not by itself establish a particular psychological, demographic, or causal explanation for that structure. Such explanations require additional analyses.

7 Threshold Selection and Sensitivity

The results support treating threshold as an analytical parameter rather than as a fixed property of the data. For the participant network, $\tau \in [0.40, 0.55]$ is a particularly informative region because the graph is sparse enough to reveal structure while still retaining a substantial connected component. For the question network, the corresponding transition occurs earlier, approximately over $\tau \in [0.30, 0.45]$.

The threshold $\tau = 0.50$ is useful for visualization because it yields a sparse, interpretable participant network and a highly selective question network. However, it should not be interpreted as a uniquely optimal threshold solely because the resulting figures are visually clear. Any substantive downstream analysis based on a particular threshold should be accompanied by sensitivity analysis over nearby thresholds.

8 Limitations

1. **Threshold dependence.** Network topology changes substantially with the correlation threshold. Conclusions based on a single threshold may therefore be sensitive to this analytical choice.
2. **Correlation is not causation.** Network edges represent statistical association between response profiles and should not be interpreted as causal relationships.
3. **Ordinal questionnaire data.** Spearman correlation is appropriate for rank-based association, but correlations between Likert responses can still be affected by response distributions, ties, and missing data.
4. **Multiple pairwise comparisons.** A large number of pairwise correlations are examined. If inferential significance testing is used in later analyses, multiple-comparison control and a minimum-overlap criterion should be considered.
5. **ER baseline limitations.** ER graphs provide a useful density-matched reference, but they do not reproduce all properties of empirical questionnaire networks. In particular, they do not explicitly model degree heterogeneity, community structure, or dependencies among correlations.
6. **Fragmented-graph metrics.** Once the network contains many disconnected components, global average path length and diameter become less informative. Interpretation in this regime should emphasize component structure rather than path-length values alone.

9 Conclusion

This analysis provides two complementary views of the questionnaire data. The participant–participant network shows a comparatively robust structure of response similarity, retaining a substantial giant component even after strong correlations are required. The question–question network is substantially more selective at high thresholds, indicating that strong item-level relationships are concentrated among a smaller subset of questionnaire items.

Both empirical networks show greater clustering than density-matched ER networks in their sparse regimes, indicating non-random local organization of similarities. The participant network additionally maintains relatively short paths within its connected portion, while the question network undergoes fragmentation earlier as the threshold increases.

Together, these results demonstrate that the questionnaire contains structured similarity at both the respondent and item levels. The next step is to connect these network structures to the substantive properties of participants and questionnaire items—for example, by examining

community membership, participant metadata, domain membership, centrality, and the stability of network communities across thresholds.

A Complete Threshold Results: Participant Network

Table 5: Complete threshold analysis for the participant–participant network.

τ	Nodes	Edges	Density	Avg. degree	Com.	Largest	Giant frac.	Clustering	Path length	Diameter
0.00	86	3655	1.000	85.00	1	86	1.000	1.000	1.000	1
0.05	86	3394	0.929	78.93	1	86	1.000	0.936	1.071	2
0.10	86	3103	0.849	72.16	1	86	1.000	0.885	1.151	2
0.15	86	2786	0.762	64.79	1	86	1.000	0.836	1.238	2
0.20	86	2375	0.650	55.23	1	86	1.000	0.769	1.351	3
0.25	86	1938	0.530	45.07	1	86	1.000	0.708	1.480	3
0.30	86	1498	0.410	34.84	1	86	1.000	0.628	1.653	4
0.35	86	1033	0.283	24.02	2	85	0.988	0.510	1.858	6
0.40	86	637	0.174	14.81	7	79	0.919	0.418	1.994	4
0.45	86	348	0.095	8.09	11	76	0.884	0.278	2.371	5
0.50	86	169	0.046	3.93	25	62	0.721	0.241	2.766	6
0.55	86	61	0.017	1.42	44	41	0.477	0.117	3.577	7
0.60	86	9	0.002	0.21	77	4	0.047	0.000	1.667	3
0.65	86	4	0.001	0.09	82	3	0.035	0.000	1.333	2
0.70	86	1	0.000	0.02	85	2	0.023	0.000	1.000	1
0.75	86	1	0.000	0.02	85	2	0.023	0.000	1.000	1
0.80	86	0	0.000	0.00	86	1	0.012	0.000	–	–
0.85	86	0	0.000	0.00	86	1	0.012	0.000	–	–
0.90	86	0	0.000	0.00	86	1	0.012	0.000	–	–

B Complete Threshold Results: Question Network

Table 6: Complete threshold analysis for the question–question network.

τ	Nodes	Edges	Density	Avg. degree	Components	Largest	Giant frac.	Clustering	Path length	Diameter
0.00	60	1770	1.000	59.00	1	60	1.000	1.000	1.000	1
0.05	60	1506	0.851	50.20	1	60	1.000	0.869	1.149	2
0.10	60	1247	0.705	41.57	1	60	1.000	0.767	1.295	2
0.15	60	972	0.549	32.40	1	60	1.000	0.671	1.452	3
0.20	60	708	0.400	23.60	1	60	1.000	0.599	1.636	3
0.25	60	501	0.283	16.70	1	60	1.000	0.502	1.922	4
0.30	60	325	0.184	10.83	4	57	0.950	0.448	2.279	5
0.35	60	186	0.105	6.20	12	49	0.817	0.310	2.643	7
0.40	60	93	0.053	3.10	20	40	0.667	0.150	2.874	8
0.45	60	40	0.023	1.33	33	23	0.383	0.094	2.838	7
0.50	60	14	0.008	0.47	47	7	0.117	0.028	2.286	5
0.55	60	8	0.005	0.27	52	5	0.083	0.000	2.000	4
0.60	60	2	0.001	0.07	58	3	0.050	0.000	1.333	2
0.65	60	1	0.001	0.03	59	2	0.033	0.000	1.000	1
0.70	60	0	0.000	0.00	60	1	0.017	0.000	–	–

τ	Nodes	Edges	Density	Avg. degree	Components	Largest	Giant frac.	Clustering	Path length	Diameter
0.75	60	0	0.000	0.00	60	1	0.017	0.000	—	—
0.80	60	0	0.000	0.00	60	1	0.017	0.000	—	—
0.85	60	0	0.000	0.00	60	1	0.017	0.000	—	—
0.90	60	0	0.000	0.00	60	1	0.017	0.000	—	—